Incremental Private Set Intersection Cardinality

Abstract

...

ACM Reference Format:
. 2026. Incremental Private Set Intersection Cardinality. In . ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

A M's zk proof

A.1 I-PSI-CA-F

Recall in Sec. **??**, we want to prove that M knows $\mathsf{tsk}_2, \mathsf{y}, r, \{\mathsf{e_d}\}_{d=0}^{n}$ such that

$$\vec{\mathsf{y}} = \mathcal{F}(\mathsf{y}) \wedge \tag{1}$$

$$\forall d \in [0, n],$$

$$\vec{c_{\mathsf{M}}}[d] = \sum_{j=0}^{\theta}{}_{\mathsf{tpk}}\ ct_i[j] +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(\vec{\mathsf{y}}_j) - 2 \times_{\mathsf{pk}_2} ct_i[j] \times_{\mathsf{pk}_2} \vec{\mathsf{y}}_j \wedge \tag{2}$$

$$c_{\vec{a}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = \mathsf{tpk}_1{}^{\mathsf{tsk}_2} \wedge \tag{3}$$

$$E = \sum_{d=0}^{n}{}_{\mathsf{tpk}}\ \mathsf{e_d} \times_{\mathsf{tpk}} \vec{a}[d] \wedge \tag{4}$$

$$\forall d \in [0, n],$$

$$\mathsf{e_d} = r \cdot \mathsf{y}^d \tag{5}$$

$$\exists\ \delta_0, \ldots, \delta_{\mathsf{n}} \in \mathbb{F}^* :$$

$$\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_{\mathsf{d}})\} \tag{6}$$

with public parameters $c_{\mathsf{M}}, E, \mathsf{tpk}, \mathsf{tpk}_1, \vec{c}_{\mathsf{T}}, \vec{a}$.

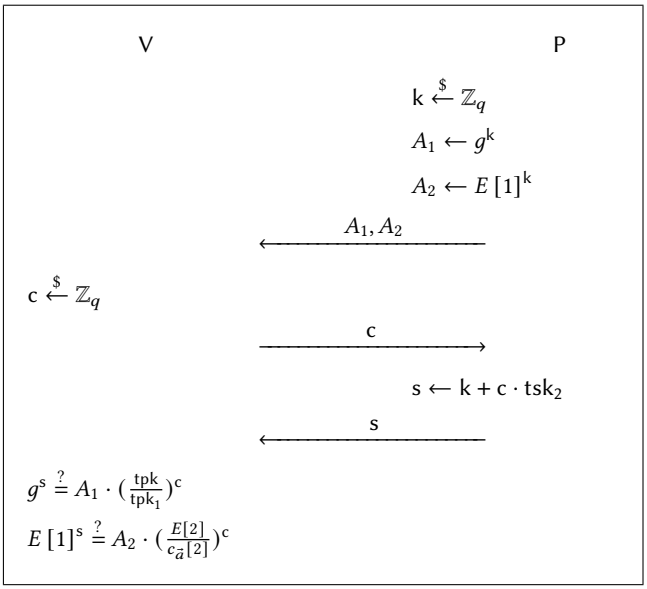


Figure 1: Proof π_{dec} for (3): $c_{\vec{a}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = \mathsf{tpk}_1{}^{\mathsf{tsk}_2}$

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference'17, Washington, DC, USA

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

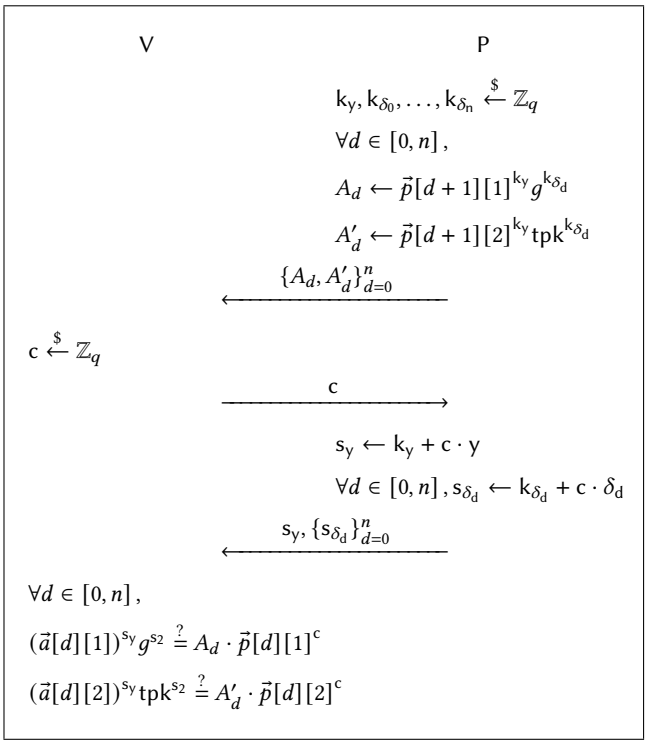


Figure 2: Proof π_{next} for (6): $\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_\mathsf{d})$

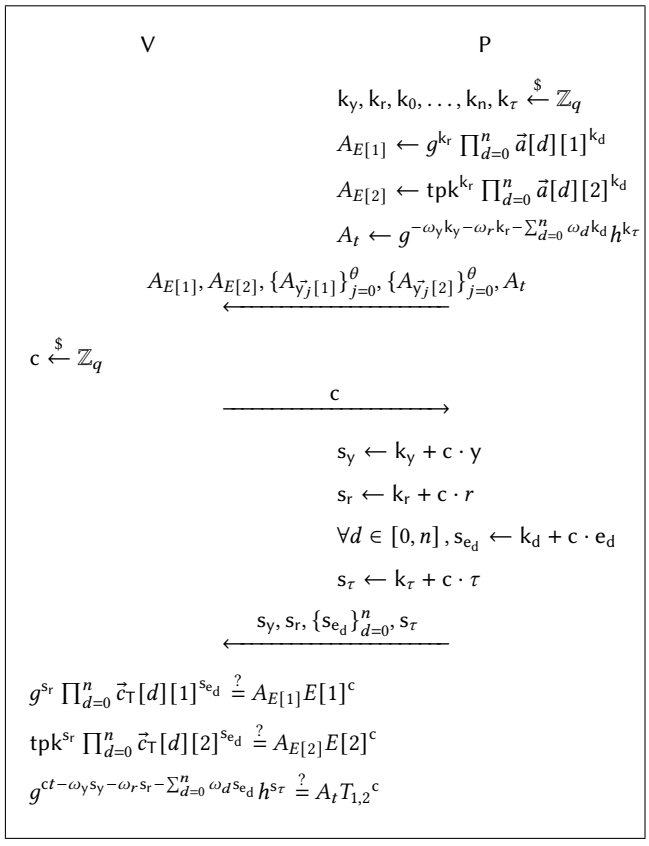


Figure 3: Proof π_Σ for Σ-Bullets Conjunction of (1), (4), (5).